

Problem Set #6: Classical Mechanics (PH31207)

October 8, 2026

Basics of Legendre Transformation

- (a) The internal energy U is a function of the entropy S , volume V , and the number of particles N . Using this and the first law of Thermodynamics $dU = TdS - PdV + \mu dN$, obtain the definition of the Gibbs free energy which should be a function of the temperature T , pressure P , and the number of particles N .
(b) What is $(\frac{\partial G}{\partial P})_{V,N}$?
2. Consider the smooth convex function $F(x)$. We define the Legendre transformation for this function as follows:

$$G(s) = sx(s) - F(x(s)),$$

where $s(x) = \frac{dF(x)}{dx}$. Now prove the following properties. Please take a look at: R. K. P. Zia, Edward F. Redish, and Susan R. McKay, “Making sense of the Legendre transform”, *Am. J. Phys.* **77**, 614–622 (2009).

- (a) Starting from $G(s(x)) + F(x) = xs(x)$, applying the chain rule, $\frac{dF}{ds} = \frac{dF}{dx} \frac{dx}{ds}$, prove that

$$\frac{dG(s)}{ds} = x(s).$$

- (b) Take the example of $F(x) = e^x$. Show that doing the Legendre transform twice gives back the original function!
- (c) Prove that

$$\left(\frac{d^2 G}{ds^2}\right) \left(\frac{d^2 F}{dx^2}\right) = 1.$$

- (d) Prove that

$$\left(\frac{d^3 G}{ds^3}\right) \left[\frac{d^2 G}{ds^2}\right]^{-3/2} + \left(\frac{d^3 F}{dx^3}\right) \left[\frac{d^2 F}{dx^2}\right]^{-3/2} = 0.$$